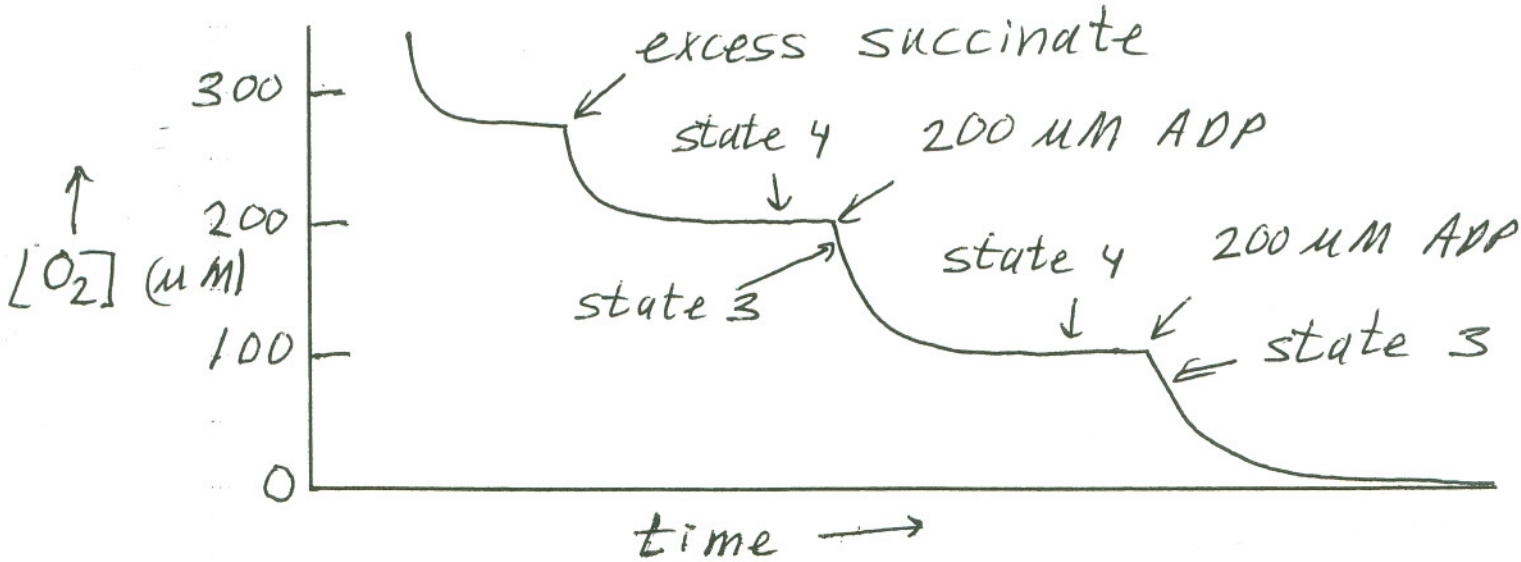
BIOCHEMISTRY CUMULATIVE EXAM
November 15, 1997

The mitochondria in a newly discovered organism have an electron transport chain that transfers electrons from succinate to O_2. (fumarate + $2H^+$ + $2e^-$ → succinate, $E_o' = -0.03$ V; $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$, $E_o' = +0.82$ V). Four redox proteins are involved in the electron transport chain, A, B, C, and D, and the mitochondria have an ATP synthase with the same properties as that in normal heart mitochondria. A suspension of the mitochondria was placed in a combined Clark O_2 electrode/spectrophotometer cell that is closed to the outside air and allows the simultaneous measurement of O_2 concentration and the redox state of each protein. Addition of excess succinate results in some consumption of O_2 as shown in the Figure, but then O_2 consumption stops, indicated by a plateau on the O_2 curve. Addition of ADP results in the consumption of additional O_2, as indicated in the Figure. The percent reduction of each protein measured in state 3 and state 4 (identified in the Figure) are given in the table below.



Biochem

Percent Reduction of Redox Protein

	A	B	C	
State 3	40%	20%	80%	60%
State 4	100%	0%	100%	100%

Questions:

1. (30) Explain what is happening at each stage of the experiment shown in the Figure, and discuss the significance of state 3 and state 4. Calculate how many moles of ATP are synthesized per mole of succinate oxidized from the above data, and explain your results

2. (50) Determine the order in which proteins A-D transport electrons from succinate to O_2 and the specific site(s) at which electron transfer is coupled to proton pumping. Draw a model for electron transport and ATP production, giving all stoichiometries. Explain your results.

3. (20) Calculate the total exergonic energy released during electron transfer from 1 mole of succinate to O_2. Discuss the efficiency with which this exergonic energy is coupled to the synthesis of ATP in these mitochondria.

   $\Delta G^{o\prime} = -RT\ln K$; $\Delta G^{o\prime} = -nF\Delta E^{o\prime}$, R = 2 cal/mol.deg, F = 23 kcal/V.mol.